FORMULARIO

Constantes

$$\varepsilon_0 = 8{,}854 \cdot 10^{-12} \text{ F/m} \quad (0.1)$$

$$\frac{1}{4\pi\varepsilon_0} = 8{,}988{\cdot}10^9 \text{ Nm}^2/\text{C}^2 \approx 9{\cdot}10^9 \text{ Nm}^2/\text{C}^2 \quad (0.2)$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} \quad (0.3)$$

$$c = 2{,}998 \cdot 10^8 \text{ m/s} \approx 3 \cdot 10^8 \text{ m/s} \quad (0.4)$$

Áreas y Volúmenes

Círculo

$$L = 2\pi r \quad (0.5)$$

$$S = \pi r^2 \quad (0.6)$$

Esfera

$$S = 4\pi r^2 \quad (0.7)$$

$$V = \frac{4}{3}\pi r^3 \quad (0.8)$$

Cilindro

$$S = 2\pi r^2 + 2\pi r L \quad (0.9)$$

$$V = \pi r^2 L \quad (0.10)$$

1 Óptica y Fotónica

Ondas

$$f(x,t) = A\sin(kx - \omega t + \phi) \quad (1.1)$$

$$\lambda = \frac{2\pi}{k} \quad (1.2)$$

$$T = \frac{1}{f} = \frac{2\pi}{\omega} \quad (1.3)$$

$$v = \lambda f = \frac{\lambda}{T} = \frac{\omega}{k} \quad (1.4)$$

Óptica geométrica

$$n = \frac{v_0}{v} \quad (1.5)$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \quad (1.6)$$

$$\sin\theta_1 = \sqrt{\left(\frac{n_2}{n_1}\right)^2 - \left(\frac{n_3}{n_1}\right)^2} \quad (1.7)$$

Óptica ondulatoria

$$\left|\vec{E}\right| = v_0 \left|\vec{B}\right| \quad (1.8)$$

$$v_0 = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad (1.9)$$

$$\vec{E}(\vec{r},t) = \vec{E}_0 \sin(\vec{k}\vec{r} - \omega t + \phi) \quad (1.10)$$

$$u_e = \frac{1}{2}\epsilon_0 E^2 \quad (1.11)$$

$$u_m = \frac{B^2}{2\mu_0} \quad (1.12)$$

$$u = u_e + u_m = \epsilon_0 E^2 = \frac{B^2}{\mu_0} \quad (1.13)$$

$$\langle u \rangle = \frac{E_0 B_0}{2\mu_0 v_0} \quad (1.14)$$

$$I = \langle u \rangle v_0 \quad (1.15)$$

$$I = \frac{E_0 B_0}{2\mu_0} \quad (1.16)$$

$$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0} \quad (1.17)$$

$$P_r = \frac{I}{v_0} \quad (1.18)$$

Difracción e interferencia

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta \quad (1.19)$$

$$\delta = k(r_1 - r_2) = k\Delta r \quad (1.20)$$

$$k = \frac{2\pi}{\lambda} \quad (1.21)$$

$$\delta = \frac{2\pi}{\lambda}\Delta r \quad (1.22)$$

Doble rendija

$$\Delta r = d \sin\theta \tag{1.23}$$

$$\delta = \frac{2\pi}{\lambda} d \sin\theta \tag{1.24}$$

Máximos:

$$d \sin\theta = m\lambda \ , \qquad m = 0, 1, 2, 3... \tag{1.25}$$

Mínimos:

$$d \sin\theta = (m - 1/2)\lambda \ , \qquad m = 1, 2, 3... \tag{1.26}$$

$$\tan\theta = \frac{y}{L} \tag{1.27}$$

Máximos:

$$y = m\frac{\lambda L}{d} \ , \qquad m = 0, 1, 2, 3... \tag{1.28}$$

Mínimos:

$$y = (m - 1/2)\frac{\lambda L}{d} \ , \qquad m = 1, 2, 3... \tag{1.29}$$

Única obertura

$$a \sin\theta = m\lambda \ , \qquad m = 1, 2, 3... \tag{1.30}$$

Óptica cuántica

$$E = hf \tag{1.31}$$

$$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s} \tag{1.32}$$

$$\lambda = \frac{h}{mv} \tag{1.33}$$

$$f = \frac{E_2 - E_1}{h} \tag{1.34}$$

Absorción estimulada:

$$\frac{dn_1}{dt} = -B_{12}\rho n_1 \tag{1.35}$$

Emisión estimulada:

$$\frac{dn_2}{dt} = -B_{21}\rho n_2 \tag{1.36}$$

Emisión espontánea:

$$\frac{dn_2}{dt} = -A_{21} n_2 \tag{1.37}$$

2 Teoría de Circuitos

$$R = \rho\frac{L}{S} \tag{2.1}$$

$$G = \frac{1}{R} \tag{2.2}$$

$$V = I \cdot R \tag{2.3}$$

Condensador

$$C = \frac{Q}{V} \tag{2.4}$$

$$i(t) = C\frac{dv(t)}{dt} \tag{2.5}$$

$$v_c(t) = \frac{1}{C}\int_{-\infty}^{t} i(t)dt \tag{2.6}$$

$$i(t) = \frac{V}{R} e^{-\frac{1}{RC}t} \tag{2.7}$$

$$v_c(t) = V(1 - e^{-\frac{1}{RC}t}) \tag{2.8}$$

$$\tau = RC \tag{2.9}$$

$$v_c(t) = V e^{-\frac{1}{RC}t} \tag{2.10}$$

Bobina

$$v_L(t) = L\frac{di(t)}{dt} \tag{2.11}$$

$$i(t) = \frac{1}{L}\int_{-\infty}^{t} v(t)dt \tag{2.12}$$

$$i(t) = \frac{V}{R}(1 - e^{-\frac{R}{L}t}) \tag{2.13}$$

$$v_L(t) = V e^{-\frac{R}{L}t} \tag{2.14}$$

$$\tau = \frac{L}{R} \tag{2.15}$$

$$i(t) = \frac{V}{R} e^{-\frac{R}{L}t} \tag{2.16}$$

3 Electrostática

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} q' \frac{(\vec{r} - \vec{r'})}{||\vec{r} - \vec{r'}||^3} = \frac{1}{4\pi\varepsilon_0} q' \frac{1}{||\vec{r} - \vec{r'}||^2} \hat{u}_{\vec{r}-\vec{r'}} \tag{3.1}$$

$$\vec{E} = \int \mathrm{d}\vec{E} = \int \frac{1}{4\pi\varepsilon_0} \mathrm{d}q' \frac{(\vec{r} - \vec{r'})}{||\vec{r} - \vec{r'}||^3} \tag{3.2}$$

$$\vec{E} = \int \mathrm{d}\vec{E} = \int \frac{1}{4\pi\varepsilon_0} \mathrm{d}q' \frac{\hat{u}_{\vec{r}-\vec{r'}}}{||\vec{r} - \vec{r'}||^2} \tag{3.3}$$

$$\mathrm{d}q' = \lambda \mathrm{d}l \tag{3.4}$$

$$\mathrm{d}q' = \sigma \mathrm{d}S \tag{3.5}$$

$$\mathrm{d}q' = \rho \mathrm{d}V \tag{3.6}$$

$$\vec{F}_e = \frac{1}{4\pi\varepsilon_0} \frac{q\,q'}{d^2} \hat{u}_{\vec{r}-\vec{r'}} = \frac{1}{4\pi\varepsilon_0} \frac{q\,q'}{d^2} \frac{(\vec{r} - \vec{r'})}{||\vec{r} - \vec{r'}||} \tag{3.7}$$

$$\vec{F}_e = q\,\vec{E}(\vec{r}) \tag{3.8}$$

$$\vec{F} = \int_\Gamma \mathrm{d}q' \vec{E}(\vec{r'}) \tag{3.9}$$

$$\phi_E = \int_S \vec{E} \cdot \mathrm{d}\vec{S} \tag{3.10}$$

$$\oint_S \vec{E} \cdot \mathrm{d}\vec{S} = \frac{Q_{int}}{\varepsilon_0} \tag{3.11}$$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0} q' \frac{1}{||\vec{r} - \vec{r'}||} \tag{3.12}$$

$$V(\vec{r}) = \int_\Omega \frac{1}{4\pi\varepsilon_0} \mathrm{d}q' \frac{1}{||\vec{r} - \vec{r'}||} \tag{3.13}$$

$$U = \frac{1}{4\pi\epsilon_0} \frac{1}{2} \sum_{i,j=1(i\neq j)}^{N} \frac{q_i q_j}{d_{ij}} \tag{3.14}$$

$$U = \frac{1}{2} \int_\Omega \mathrm{d}q' V(\vec{r'}) = \frac{\varepsilon_0}{2} \int_T ||\vec{E}(\vec{r})||^2 \mathrm{d}^3\vec{r}, \tag{3.15}$$

$$E = \frac{\sigma}{\epsilon_0} \tag{3.16}$$

$$|\,\Delta V\,| = \frac{\sigma}{\epsilon_0} d = \frac{Q}{\epsilon_0\,S} d \tag{3.17}$$

$$C = \frac{Q}{|\,\Delta V\,|} = \frac{\epsilon_0}{d} S \tag{3.18}$$

4 Magnetostática e Inducción electromagnética

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times (\vec{r} - \vec{r'})}{||\vec{r} - \vec{r'}||^3} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{u}_{\vec{r}-\vec{r'}}}{||\vec{r} - \vec{r'}||^2} \tag{4.1}$$

$$\vec{B} = \int_\mathcal{C} \frac{\mu_0}{4\pi} \frac{I\mathrm{d}\vec{l} \times (\vec{r} - \vec{r'})}{||\vec{r} - \vec{r'}||^3} = \int_\mathcal{C} \frac{\mu_0}{4\pi} \frac{I\mathrm{d}\vec{l} \times \hat{u}_{\vec{r}-\vec{r'}}}{||\vec{r} - \vec{r'}||^2} \tag{4.2}$$

$$\oint_\Gamma \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 I_{int} \tag{4.3}$$

$$\vec{F}_{1\to 2} = \frac{\mu_0}{4\pi} \frac{q_1 q_2}{d^2} (\vec{v}_2 \times (\vec{v}_1 \times \hat{u}_{\vec{r_2}-\vec{r_1}})) \tag{4.4}$$

$$\vec{F}_m = q\vec{v} \times \vec{B}(\vec{r}) \tag{4.5}$$

$$\vec{F}_m = \int_\Gamma I\mathrm{d}\vec{l} \times \vec{B}(\vec{r}) \tag{4.6}$$

$$\vec{F}_{em} = q\left[\vec{E}(\vec{r}) + \vec{v} \times \vec{B}(\vec{r})\right] \tag{4.7}$$

$$\phi_M = \int_S \vec{B} \cdot \mathrm{d}\vec{S} \tag{4.8}$$

$$\mathrm{f.e.m} = -\frac{\mathrm{d}\phi_B}{\mathrm{d}t} \tag{4.9}$$

$$B = \mu_0 n I \tag{4.10}$$

$$L = \frac{d\Phi}{dI} \tag{4.11}$$

$$\varepsilon = -L\frac{dI}{dt} \tag{4.12}$$

5 Semiconductores

Los materiales semiconductor3ss

$$E_n = \frac{-13.6}{n^2}\ \text{eV} \tag{5.1}$$

$$n_\mathrm{i} = N_C \cdot e^{-\frac{E_C - E_F}{k_B T}} \tag{5.2}$$

$$p_\mathrm{i} = N_V \cdot e^{-\frac{E_F - E_V}{k_B T}} \tag{5.3}$$

$$E_F = \frac{E_C + E_V}{2} + \frac{3k_B T}{4} \ln\left(\frac{m_p}{m_n}\right). \tag{5.4}$$

$$n \cdot p = n_\mathrm{i}^2 \tag{5.5}$$

$$E_{F,\mathrm{n}} = E_{F,\mathrm{i}} + k_B T \cdot \ln\left(\frac{N_D}{n_\mathrm{i}}\right) \tag{5.6}$$

$$n_\mathrm{n} = N_D,\ p_\mathrm{n} = \frac{n_\mathrm{i}^2}{N_D} \tag{5.7}$$

$$E_{F,\mathrm{p}} = E_{F,\mathrm{i}} - k_B T \cdot \ln\left(\frac{N_A}{n_\mathrm{i}}\right) \tag{5.8}$$

$$p_\mathrm{p} = N_A,\ n_\mathrm{p} = \frac{n_\mathrm{i}^2}{N_A} \tag{5.9}$$

$$\vec{v}_{d,n} = -\mu_n \cdot \vec{E} \tag{5.10}$$

$$\vec{v}_{d,p} = \mu_p \cdot \vec{E} \tag{5.11}$$

$$I = \frac{V}{\left[\frac{1}{q(\mu_p \cdot p + \mu_n \cdot n)}\right]\frac{L}{A}} = \frac{V}{R} \tag{5.12}$$

$$I = \frac{\Delta Q}{\tau_c} = (n_2 - n_1)\frac{q \cdot l}{2\tau_c}A = (n_2 - n_1)\frac{q \cdot v_{th}}{2}A \tag{5.13}$$

La unión p-n. Los díodos

$$V_{bi} = \frac{1}{2}\frac{q\,(N_D x_\mathrm{n} + N_A x_\mathrm{p})}{2\epsilon_0\epsilon_r}W \tag{5.14}$$

$$V_{bi} = \frac{k_B T}{q}\ln\left(\frac{N_A \cdot N_D}{n_\mathrm{i}^2}\right) \tag{5.15}$$

$$W = \left[\frac{2\epsilon_0\epsilon_r V_{bi}}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\right]^{1/2} \tag{5.16}$$

$$x_\mathrm{p} = W \cdot \frac{N_D}{N_A + N_D} \tag{5.17}$$

$$x_\mathrm{n} = W \cdot \frac{N_A}{N_A + N_D} \tag{5.18}$$

$$n(x = -x_\mathrm{p}) = n_{\mathrm{p},0} \cdot e^{\frac{qV}{k_B T}} \tag{5.19}$$

$$p(x = x_\mathrm{n}) = p_{\mathrm{n},0} \cdot e^{\frac{qV}{k_B T}} \tag{5.20}$$

$$I(V) = I_s\left(e^{q \cdot V / k_B T} - 1\right) \tag{5.21}$$